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Design process of a lounge table

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Abstract

This thesis investigates the design and development of a lounge table made exclusively from damaged spruce wood, with the ambition to explore sustainable and meaningful furniture design within an industrial context. The project addresses the central research question: *How can a lounge table be designed to embody meaning while balancing desirability, feasibility, viability, and sustainability?* In collaboration with the design company Sense By Us, a design process was initiated that followed a human-centered and iterative structure. The work integrates theoretical perspectives from industrial design, design thinking, product development, perception theory, and sustainability frameworks.

The project began with research and ideation, followed by concept development, 3D modeling, physical mock-ups, and ultimately the production of a full-scale prototype. The design process placed high emphasis on material limitations, aiming to elevate the aesthetic and structural value of wood that would otherwise be considered waste. Throughout the development stages, reflection-in-action was applied to guide decisions and evaluate progress, combining practical craftsmanship with design theory.

The result is a lounge table that merges a sculptural organic wooden base with a contrasting oval glass top, expressing a harmony between form, material, and function. While the prototype successfully embodies the project's values and aesthetic goals, full validation with users could not be carried out due to time constraints and external limitations. Nevertheless, the thesis outlines a clear path for future validation methods, such as focus groups and collaboration with interior professionals. The work contributes to the field of industrial and sustainable design by demonstrating how damaged materials can be reimagined into high-value products through a rigorous and reflective design process.

Sammanfattning

Detta examensarbete undersöker design och utveckling av ett soffbord tillverkat uteslutande av skadat granvirke, med målet att utforska hållbar och meningsfull möbeldesign inom en industriell kontext. Projektet adresserar den centrala forskningsfrågan: *Hur kan ett soffbord formges på ett meningsfullt sätt med hänsyn till önskvärdhet, genomförbarhet, lönsamhet och hållbarhet?* I samarbete med designföretaget Sense By Us inleddes en designprocess som följde en mänskligt centrerad och iterativ struktur. Arbetet integrerar teoretiska perspektiv från industridesign, design thinking, produktutveckling, perceptionsteori och hållbarhetsramverk.

Projektet inleddes med research och idéarbete, följt av konceptutveckling, 3D-modellering, fysiska modeller och slutligen framställning av en fullskalig prototyp. Stor vikt lades vid materialets begränsningar, med ambitionen att lyfta estetiska och strukturella värden i trä som annars skulle betraktas som avfall. Genom hela utvecklingsprocessen tillämpades reflektion i handling för att styra beslut och utvärdera arbetets framsteg, där hantverk kombinerades med designteori.

Resultatet är ett soffbord som kombinerar en skulptural organisk träbas med en kontrasterande oval glasskiva, vilket uttrycker harmoni mellan form, material och funktion. Även om prototypen framgångsrikt förmedlar projektets värden och estetiska mål kunde en fullständig användarvalidering inte genomföras på grund av tidsbrist och externa begränsningar. Arbetet redovisar dock tydliga möjligheter för framtida valideringsmetoder, såsom fokusgrupper och samverkan med inredningsproffs. Examensarbetet bidrar till fältet för industriell och hållbar design genom att visa hur skadat material kan omformas till produkter med högt värde genom en reflekterande och metodisk designprocess.

Table of Contents

1 Introduction 1
1.1 Company information 3
1.2 Objective and purpose 3
1.3 Scope and limitations 3

2 Theoretical framework 4
2.1 Product development 4
2.2 Industrial design 4
2.3 Design for sustainability 5
2.4 The Double Diamond Framework 5
2.5 Design philosophy 7
2.5.1 Reflection in action 7
2.5.2 Sustainable design 8
2.5.3 Golden circle 8
2.6 Sensation perception 9
2.6.1 Empathy map 10
2.6.2 Why we love and hate everyday things 11

3 METHODS 13
3.1 Double diamond methodology 13
3.2 Discover 13
3.2.1 Market research 13
3.3 Define 13
3.3.1 Persona 13
3.3.2 Moodboard 13
3.3.3 Requirement specification 13
3.4 Develop 14
3.4.1 Sketching 14
3.4.2 Concept evaluation 14
3.4.3 Brainstorming 14
3.4.4 Mockup 14
3.4.5 CAD 14
3.4.6 Rendering 14
3.5 Deliver 15
3.5.1 Prototype 15

4 Implementation & Approaches 16
4.1 Discover 16
4.1.1 Market research 16
4.2 Define 16
4.2.1 Personas 16
4.2.2 Moodboard 18
4.2.3 Requirements specification 18
4.3 Develop 19
4.3.1 First iteration Sketches 19
4.3.1.1 Second iteration sketches 20
4.3.2 CAD 20
4.3.3 Mock-up 21
4.3.4 Rendering 22

4.3.5 Concept evaluation 22
5 Result 24
5.1 Result of research 24
5.1.1 Sustainability 24
5.1.2 Desirability 24
5.1.3 Feasibility 24
5.1.4 Viability 25
5.2 Result of design proposal 25
5.2.1 Shape and Expression 25
5.2.2 Tabletop 25
5.2.3 Attachment between tabletop and base 26
5.2.4 CAD 27
5.2.5 E-drawing 27
5.2.6 Render 27
5.2.7 Contextual render 28
5.2.8 Prototype 29
6 Discussion and conclusions 33
6.1 Discussion on research questions 33
6.1.1 How to design a meaningful lounge table considering desirability, viability, sustainability and feasibility? 33
6.1.2 What is a meaningful product? 33
6.1.3 The role of storytelling 34
6.1.4 Contribution to the field 34
6.2 Conclusions and future scope 35
6.2.1 Future scope 35
6.2.2 Validation considerations 36
7 Reference list 37

1 Introduction

This thesis is on furniture design as part of industrial design, with a focus on the design of a lounge table. The goal is to discover how product design principles of desirability, viability, sustainability, and feasibility can be applied to transform low-value or abandoned materials into high-quality and significant furniture.

Furniture design, when addressed from a design-led stance, involves more than the creation of aesthetically pleasing objects. It requires insight into user behavior, the characteristics of materials, manufacturing processes, and environmental responsibility. In this project, European spruce bark beetle-infested spruce wood has been chosen as the primary material. This wood, which would typically be considered unsuitable for high-quality applications, presents practical as well as conceptual challenges that are explored through physical experimentation and iterative development.

Industrial design demands both problem-solving and creative thinking. It demands solutions that are not only cost-effective and functional but also emotionally relevant and culturally suitable. The thesis demonstrates how systematic design processes such as the Double Diamond model and empathy mapping can contribute to the creation of products that reflect dominant values related to sustainability and material narrative.

As sustainability determines the future of design, this project explores how material decision-making and narrative communication can increase the perceived value of a product. By recontextualizing an imperfect or waste material as a desirable object, the project proposes an alternative mode of consumption, resource use, and messaging through furniture design.

Since 2004, annual dead wood in forests has varied between 8 and 13 million m³. While natural factors such as fire, wind, and snow have historically played a role, spruce bark beetles have increasingly become the prevailing cause. Extreme weather, such as the 2018 drought and severe storms, has helped to enable the movement of beetles, leading to increased dead wood in forests. Human activities also worsen the situation. Tapping trees leaves up to 1 million m³ of dead timber each year, and forestry activities including thinning and regeneration harvesting subject forests to greater risk of wind and snow damage, which in turn benefit spruce bark beetle infestations. The amount of dead wood due to different reasons, the majority of which was due to spruce bark beetles, rose from about 2 million m³ to almost 5 million m³ per year. This shows the growing effect of spruce bark beetle infestation on forest ecosystems. for 28 seconds

Additionally, dead wood caused by other or unknown causes, primarily due to the infestation of spruce bark beetle, has grown from somewhat over 2 million m³ to nearly 5 million m³ per year. This path, combined with contributions from natural tree death and decomposition, shows the growing function of spruce bark beetle outbreaks in the forest ecosystem [1].

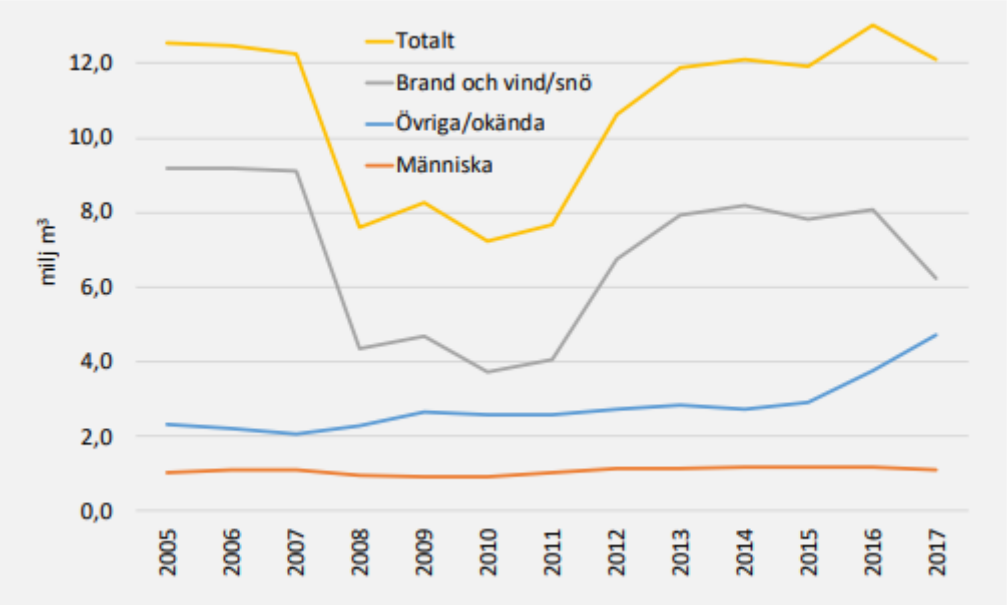


Figure 1 Annual Volume of Dead Wood in Swedish Forests

Storm-destroyed forests have the potential for a great degree of damage through such means as increasing the expansion of the spruce bark beetle. The spruce bark beetle inhabits Sweden nationally and is under the bark in recently deceased spruce where they reproduce and thus lower wood quality and even bring about mortality for the worst-affected trees. The greater the number of trees destroyed, the more space is available for colonizing beetles and thus more devastation over time. To remove such a threat, owners of forest land must quickly harvest and remove storm-damaged trees. Weather also comes into the picture as a determinant since beneficial temperature and warm weather conditions of spring and summer seasons trigger the spruce bark beetle populations. The magnitude of threat through infestation also depends upon forest structure.

The Ministry of Finance in March 2005 announced financial assistance to forest owners whose forests were destroyed by the storm. They were paid at the rate of 50 SEK per cubic meter of storm-killed forest. The aim was to motivate forest owners to remove fallen timber as soon as possible because it exposed more forests to infestation by spruce bark beetles. The tax reduction was based on the amount of wood (cubic meters solid under bark) exported from the forest and equaled 2 billion SEK for the years 2005 and 2006 [2].

Spruce bark beetle timber is used as building wood and is a product which is not visible in finished buildings. Since the timber does not have compromised strength properties, it could theoretically be used as usual timber. However, its use is limited in certain markets, primarily for beauty reasons.

Demand for spruce bark beetle timber, i.e., blue stain-infested timber, varies across countries. Blue stain is better tolerated in the United States, at least in part due to familiarity with color shifts in wood. Some of the reason is that U.S. consumers purchase enormous quantities of timber from Canada, where much of the logging is in winter. The timber is then stored under snow and sawdust before processing, resulting in natural color shifts to which the U.S. marketplace has become tolerant.
In Sweden and Japan, blue stain is not so favored, although it does not compromise the timber. One reason for this is that there is no traditional practice of using blue-stained timber, and thus people perceive that it is low-quality timber. Blue stains are also misconceived by many consumers as mold, though blue stain fungi are eliminated in drying in sawmills and result in discoloration only.

A particular issue is faced in treating sideboards, which show blue stain very quickly. They have a smaller market and are therefore used almost exclusively for packaging. Even in this case, however, the issue is cosmetic, as the functional characteristics of the boards are not impacted [3].

Spruce bark beetle infestations reduce the quality of timber by making the bark peel off when infested and a spruce tree becomes part of the dying process. The removal of bark exposes the timber to a blue stain, and storage rots. The timber is therefore no longer graded as pulpwood or saw timber, but must be sold as firewood, a much lower marketable product [4].

Beyond the obvious environmental concerns, there's another issue that's been quietly piling up: large amounts of beetle-damaged timber that aren't being put to good use. But what if that's not just a problem, what if it's an opportunity. In a world increasingly focused on sustainability and circular economies, it's worth asking whether some of the materials we've written off as waste could become something meaningful.

The real challenge is learning to see these flawed materials in a new light, not just as functional, but as something that people might want. There is a design movement now that requires more than function or form, people want objects that convey values, start conversations, and feel significant. Design, in this sense, becomes a kind of storytelling.

Design thinkers like Donald Schön have cited that the best designers don't necessarily start off with clearly defined problems or solutions. Instead, they stumble about getting things along the way, and the process of design informs the outcome [5]. On another but related note, Simon Sinek's Golden Circle principle instructs that the most impactful brands and products are those that start with "why" why they exist, and why everyone cares. It's not just about making something that works; it's about making something that speaks [6].

That type of thinking is especially relevant in environments like interior and workplace design, where stakes are no longer just aesthetic, ethical. Organizations are increasingly being led by purchase decisions made with sustainability goals in mind, particularly in B2B setting. That's the world this project is entering. The goal is to

take beetle-killed spruce, something usually relegated to scrap and make it into functional furniture that's full of meaning [7].

1.1 Company information

Sense By Us has deep industry experience in the manufacturing of furniture and interiors, which makes the company well-placed to meet various customer needs and wants. It has been clear over the years that concentration on core competencies, while covering areas of lesser proficiency by way of strategic partnerships, brings immense advantages. Consequently, the company prefers close cooperation with an extensive network of suppliers, a practice that not only ensures excellent product quality but also competitive pricing using the most appropriate supplier for each specific component.

Furthermore, by working in cooperation with a wide variety of materials, Sense By Us can provide customers with total interior solutions from a single source. This centralized approach guarantees that both design integrity and functionality are tackled in the best possible manner. Clients have the advantage of a simplified process with one company and one contact person throughout the project. This not only enhances convenience and operational efficiency but also results in significant time savings. The smooth workflow right from the initial stages of project conceptualization to the final product delivery also enables cost efficiency. Partnership, in fact, remains a cornerstone of the company's philosophy, with a commitment to working closely with clients from the inception of a project through to the achievement of its objectives [8].

1.2 Objective and purpose

The main objective of this thesis is to develop a design concept for a lounge table using only damaged spruce wood. The goal is to incorporate the key principles of desirability, viability, sustainability, and feasibility into the design, ensuring it meets the diverse needs and expectations of a lounge table.
This research aims to answer the following questions:

- How can a meaningful lounge table be designed considering desirability, viability, sustainability, and feasibility?
- What makes a product meaningful?

1.3 Scope and limitations

This thesis deals with the development of a lounge table design based on sustainable material selection, that is from spruce wood infected with the European spruce bark beetle. The scope entails an end-to-end design process grounded on desirability, feasibility, viability, and sustainability principles. Key activities include user research, market study, concept development, prototyping, and evaluation, and special emphasis on storytelling and ecological value as part of product design.

The design process is professional-space-specific, such as office waiting rooms and lounges, answering the current interior trends on minimalism, natural aesthetics, and eco-friendly material acquisition. Although the primary material focus is beetle-killed spruce, secondary materials such as glass are investigated to foster functionality and design minimalism.

Still, the project has some limitations. It does not entail mass production analysis or complete cost calculations beyond prototyping on small scales. Structural stress testing or long-term endurance testing is outside the scope also. The study limits itself to a Scandinavian market and design situation that may impact the generalizability of outcomes in other cultural or commercial contexts. Furthermore, the prototype is constructed using in-house manufacturing equipment, which can be an underestimate of more widespread industrial manufacturing capacity.

Despite these restrictions, the project aims to offer insightful observations of sustainable furniture design and explore the extent to which powerful narratives can be embedded in mundane objects through deliberate material and aesthetic choices.

2 Theoretical framework

This chapter discusses the theoretical basis upon which the final product in this thesis was created and outlines the principal academic and professional perspectives encompassing the design process, specifically product development, industrial design, design thinking, and perceptual experience.

Referring to the existing theory and models, the chapter aims to outline the way design disciplines manage issues related to functioning, sustainability, aesthetics and user interaction. Increasing understanding of the principles, models, and philosophy applicable in the design of user-focused, meaningful products is established in each subsequent section. Such theoretical understandings form the tools through which the problem being discussed is analyzed and a suitable design solution created.

2.1 Product development

Product development is a process where each phase advances a concept towards a product that gets launched. It begins with the need and opportunity identification of the user and concept generation and evaluation and product architecture development. Each stage plays a key role in influencing the development of the product's function, form, and feasibility.

This is a process intrinsically multidisciplinary and requires tight cross-functioning across disciplines such as design, marketing, and engineering. Cross-functioning renders the product technologically feasible and inexpensive to produce while fulfilling customer needs.

Development stages include system design at the system level, detailed design, prototyping, and testing and refining to deliver production-quality output. The processes are usually iterative and cyclical with feedback from testing to design to iteratively optimize and hone the product over time.

Product development in this scenario attempts to strike a balance across the three critical parameters of quality, time-to-market, and cost. Quality product development is achievable only through a controlled process and awareness about market and user needs as well as with proper management of the project. The customer focus does not just mean the product is working but also creating actual value to the users [9].

2.2 Industrial design

Industrial design involves the integration of creativity, form, and producibility to provide products that satisfy both the user's needs and market needs. Industrial designing is the intentional designing of products to be produced in larger numbers, balancing form and usability to devise solutions that work and are logical.

Industrial design essentially involves the way things look, how they are experienced, and how they function. It involves all the elements that range from form and material to user experience and manufacturing process. According to the Industrial Designers Society of America, industrial design involves the professional process of designing products that people use millions of times a day all over the world. This highlights the point that the look of the product alone is not important but also how it functions, how it will be made, and what user experience it will provide for people.

Industrial design is also a bridge between human need and technological potential. It must collaborate interdisciplinarity with engineering, marketing, and commerce to achieve the compatibility of the product to the name identity of the brand, sustainability, and the production limitations. According to Charles Eames, design is a blueprint for putting things in the most effective order to perform a given function, emphasizing the intentionality of what goes into the work and how it influences people and the world. The roots of industrial design are found in the early century movements of the Deutscher Werkbund and the Bauhaus that grappled to reconcile industry, craft, and art. It was these movements that clarified not only the concept of design as a

specialist discipline distinct from decoration but as a cultural and functional practice for the improvement of society. It is this heritage that continues to influence industrial design in terms of changing values as much as the modern need for sustainability for practice.

In the context of this thesis, industrial design provides the theoretical and practical foundation for designing a lounge table that not only focuses on user experience and market suitability but also on material sustainability and producibility. This concurs with the modern expectations that design interacts directly with circular economies and sustainable consumption using existing materials such as beetle-killed spruce timber and the production of products of durable value [10].

2.3 Design for sustainability

Design thinking is a cyclical human centered process that emphasizes empathy, creativity, and logic in solving problems by considering the needs of users, technological viability, and business success. Design thinking has three fundamental pillars at its core: desirability, feasibility, and viability.

Desirability is concerned with understanding what people require and desire. It is concerned with gaining a deep insight into people's behavior, emotions, and areas of pain. This is achieved through methods such as interviews, user observation, and plunging research. It is to ensure that solutions developed are meaningful and enable people to relate to them at an everyday level.

Feasibility is the technical and operational feasibility of a solution being proposed. It is the process of assessing whether an idea can be done using the tools, technology, and capability at hand. This stage decides that ideas not only have to be creative but also feasible to implement. Designers collaborate with engineers, developers, and production teams to get creative visions aligned with what can be produced.

Viability refers to the money-related viability of the solution. It examines if the idea can be of use and viable in a business setting. This encompasses estimating costs, potential revenue, market demand, as well as strategic fit. A solution must be financially viable for it to remain sustainable in the long term.

For a solution to be truly sustainable, all three principles must be addressed together. A product that is desirable but not feasible cannot be built. A product that is feasible but not viable will not last. And a product that is viable but not desirable will fail to attract users. The strength of design thinking lies in balancing these dimensions to create solutions that are innovative, implementable, and lasting.

By blending empathy and testing, and the collaboration of human instinct with strategic planning, design thinking enables responsible and inclusive innovation. Design thinking calls on designers to transcend thoughts of form or bare function and consider the ways in which their solutions affect people, society, and the world. With this system's perspective comes the result of not only effective but also ethical and sustainable solutions [11].

2.4 The Double Diamond Framework

Double Diamond, one of the most popular models in design and product management, was designed to introduce structure and focus to the typically iterative and poorly defined process of design. Its origin dates to 2005 when the UK Design Council desired to eliminate the challenges of dealing with complicated design projects by creating a system that graphically and methodologically describes the process from problem identification to solution implementation. Double Diamond model was created from extensive research and consultation with the best design industry professionals with the goal of distilling the key activities and cognitive processes that produce successful design outcomes. By encapsulating these discoveries in a valuable model, the UK Design Council offered a powerful tool that enriches knowledge about the design process while facilitating collaboration and communication for cross-functional teams. Since its inception, the framework has undergone development as the central theory of design thinking and product management with a universal applicability for crossing the journey from idea to rollout.

The Double Diamond process is an image description of the design journey, broken down into two intersecting diamonds, that symbolizes the double process of divergent and convergent thinking which is necessary to be effective in design practice. Every diamond symbolizes a particular stage in the process: the first encloses the Discover and Define stages, and the second encloses the Develop and Deliver stages. This separation highlights

the nature of design being cyclical in that exploration and refinement occur in cycles, and not through a single linear progression.

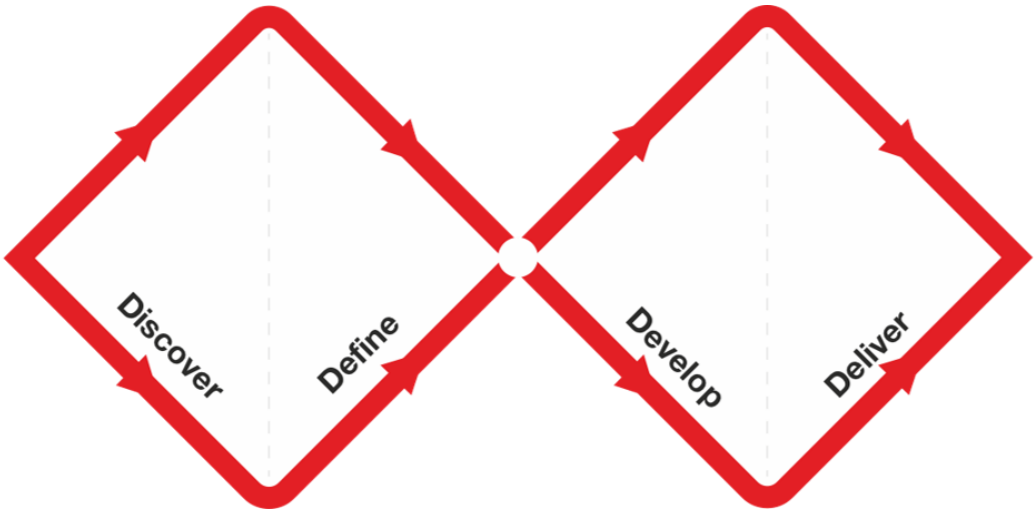


Figure 2 The double diamond design process

Though originally designed as part of the design process, the Double Diamond model has subsequently been adapted effectively into product management to provide a systematic yet fluid road map for the product development stages. The four phases fit the important activities by which the product managers may move the complexity of having a concept into a product ready to hit the market. By ensuring intensive examination, validation, and optimization at every stage, this process improves the likelihood of creating a successful and market-aware product.

Discover Phase: Finding Out About Market and User Requirements

This phase is designed to develop an understanding by gathering insights, establishing user needs, and exploring the problem space in depth. Characterized by divergent thinking, this phase encourages exploration of several perspectives and collection of broad data [12].

The Discover phase, the broadest in scope, involves gaining thorough understanding of the market, user activity, and the broader problem space. Product managers can draw on several research methods like market research, user interviewing, and competitive benchmarking in a bid to collect data to inform the next decisions. Divergent thinking defines this stage, encouraging loose exploration of future user needs, pain points, and market spaces. Deliverables at this stage often include persona development, customer journey maps, and research summaries that collate observations into usable implications [13].

Define Phase: Refining the Problem Statement

In this phase, the process shifts from divergence to convergence, where information gathered is consolidated to develop a concise and actionable problem statement or design brief. This phase determines the direction of the project, such that all further work will be grounded in needs and objectives identified [12].

After comprehensive research carried out during the Discover phase, the Define phase transforms into a convergent phase by narrowing down information gathered into an articulate and coherent problem statement. The Define phase is important to narrow down wide possibilities into focused objectives that would be utilized for guiding product development activities. Product managers set a strategic vision, both describing the problem to be solved and the value proposition of the product. Define will often conclude with the creation of a detailed product brief or roadmap, aligning stakeholders around key features, target markets, and business objectives [13].

Develop Phase: Ideation and Prototyping

Labeled as a transition from problem space to solution space, the develop phase re-introduces divergent thinking to explore multiple possibilities for solutions. Ideation, prototyping, and testing become key activities allowing the sharpening of design concepts against feedback and assessment [12].

The transition from Define to Develop is the shift from problem analysis to solution exploration. This phase is characterized by prototyping, ideation, and continuous testing, all towards the goal of iterating and testing potential solutions. Product managers collaborate extensively with designers, engineers, and cross-functional teams to generate multiple concepts and experiments through rapid iterations. User testing and feedback form the heart of this phase since they provide ongoing optimization based on empirical data. By the end of the Develop phase, the product concept is solidified and finalized, with a clear idea of its strengths and weaknesses [13].

Deliver Phase: Implementation and Market Launch

The final stage finishes the process by focusing on convergent thought, formulating and implementing the chosen solution. Testing, finish, and deploy the product come first, in addition to upholding standards and user needs previously set [12].

The Deliver phase is the final phase of the product development process. During this phase, the focus is no longer on iterative refinement but on production completion and market readiness. Product managers oversee aggressive testing, like quality assurance, user acceptance testing, and beta testing, to ensure that the product meets market requirements and user expectations. This stage also involves creating a go-to-market strategy, including marketing, sales, customer support, and post-launch analysis. The success of this stage is measured by the market acceptance of the product, user satisfaction, and business goals alignment.
By integrating the Double Diamond model in product management, teams have a systematic approach to tackling the intricacies of product development. Its emphasis on iterative exploration and iteration makes products not only innovative but also user-centric and market-ready [13].

2.5 Design philosophy

Design philosophy entails the value system, intent, and approach that drives how designers handle creative challenges. For Brian Lawson in How Designers Think, design is not a process of reacting to well-defined problems but addressing messy, ill-defined problems that necessitate analysis as well as creative instinct. Problems in design, he says, are fundamentally "ill-defined" or "wicked," have no one correct solution, and need reformulation numerous times along the route via the process. Rather than being rational or linear, the design process entails circularity and reflection guided by constraints, user requirements, and the designer's personal judgment. Lawson also states that design thinking entails a mix of creativity and reason, and that designers must balance aesthetic, functional, technical, and moral factors. This process is a central theme of most modern design philosophy whereby the generation of meaningful, sustainable, and context-specific solutions is privileged over technical optimization for its own sake [14].

2.5.1 Reflection in action

In The Reflective Practitioner (1983) Donald A. Schön offers a background perspective on professional practice, especially in design, architecture, and other creative professions. Schön dissolves the widespread stereotype of the professional as merely a technical problem solver using known knowledge to well-defined problems. Instead, he proposes that much of real practice is in uncertainty, instability, uniqueness, and value conflict settings, environments where conventional problem-solving methods are insufficient.

Central to Schön's theory is the idea of "reflection in action" that describes the way experts think critically and creatively in the middle of practice. Unlike a sequential design process, practitioners develop a kind of dialogue with the situation in which every step leads to a new awareness and feeds forward into the next. This looping, experimental mode of working supports real-time reformulation and reframing of the problem.

It is as crucial as "reflection on action" coming after the activity has been done. In this, practitioners think about and examine their decisions, achievements, and failures in developing learnings to apply to upcoming

projects. It helps in retrospect with ongoing learning and the development of professional instinct and judgment.

Schön's model also stresses the tacit knowledge, that is the non-verbal, experiential knowledge which underlies much of a designer's practice. By prioritizing reflective practice over methodologies where strict rules apply, his work provides an explanation of how designers can cope with complexity, ambiguity, and changing constraints.

It has become influential in design education and research, providing a framework for the interpretation of design as a reflexive, adaptive, and context-dependent practice. It emphasizes the importance of flexibility, critique, and ongoing refinement of professional identity through experience [5].

2.5.2 Sustainable design

Sven Lundh, as both a businessman and as a producer, is defined in Personakt as an artist's representative rather than an artist in his own right. One of the ongoing concerns in his practice was that an item must "stand wear and tear on the eye.". This is a theme repeated by Sune Nordgren in his book and one that summarizes Lundh's design and philosophical stance. In other words, an item must not only function or be fashionable at a given moment, or at least stylistically modern, but also has staying power in its materiality as well as its looks.

We can gather from Rebecka Tarschys's text that Lundh never strove for "pleasant" or "commercial success" but only for something that would engage and challenge the spectator. With his activity at Källemo, he was able to create a platform where artistic activity would have space without sacrificing its idea or its integrity.

Gunnar Lindqvist portrays Lundh's appreciation for quality and trust in artists' expression. It is this faith that made cooperation with designers such as Mats Theselius and John Kandell possible. It is also here that the ideal is subjected to wear and tear to the eye with a practical use – not only must it endure use but also appear aesthetically modern over the years.

Based on these texts, Lundh's theory was firmly rooted in an idea-based theory of the object, rather than fashion or trends. He used terms such as "idea-based", "non-pragmatic production" and "intellectual furniture" – terms used in several of the book's texts and terms that indicate his desire to position himself outside aesthetically normal [15].

”Tänker man efter är det inte självklart att kvalitet bara kan mätas i form av ergonomiska mått, ytors tålighet och limfogars hållbarhet. Kvalitet är ett vidare begrepp i hög grad har att göra med ögats erfarenhet. Att vara giltig längre är god kvalitet. Då förlänger man vägen till containern.”

Citat av Sven Lundh (Nordgren 2007, s. 160)

Translation

“When you think about it, it's not obvious that quality can only be measured in terms of ergonomic standards, surface durability, and the strength of glued joints. Quality is a broader concept that has a lot to do with the eye's experience. Lasting validity is good quality. That way, the road to the dumpster is extended.”

2.5.3 Golden circle

The Golden Circle is a leadership and communication model devised by Simon Sinek that explores how people and businesses can motivate action and commitment by reversing the normal order of communication. The model is made up of three concentric circles: WHY, HOW, and WHAT.

- WHY is the core motivation, reason, or objective that inspires an organization or someone. It is an underlying reason for existing behind building and sustaining an organization. It is not bottom-line goals or profit, but rather something deeper that guides action and vision.

- HOW addresses what mechanisms, values, or principles guide how WHY is accomplished. It involves the specific way or fashion through which an entity is distinguished from others.
- WHAT is something physical that an organization or a person produces, such as a product, a service, or an outcome.

Sinek says that most companies talk from the outside in, beginning with WHAT they do and how they do it, with little mention of WHY. The most successful and long-standing businesses and leaders talk from the inside out, beginning with WHY. By doing this, they connect more on an emotional level with people and build greater trust and loyalty.

The Golden Circle framework is widely utilized across different areas including leadership, marketing, branding, and organizational development. It is utilized as a strategic instrument to ensure that actions and communications are aligned with defined WHY. By beginning with WHY, organizations can inspire rather than manipulate people's actions, leading to more long-term engagement and sustainable success [16].

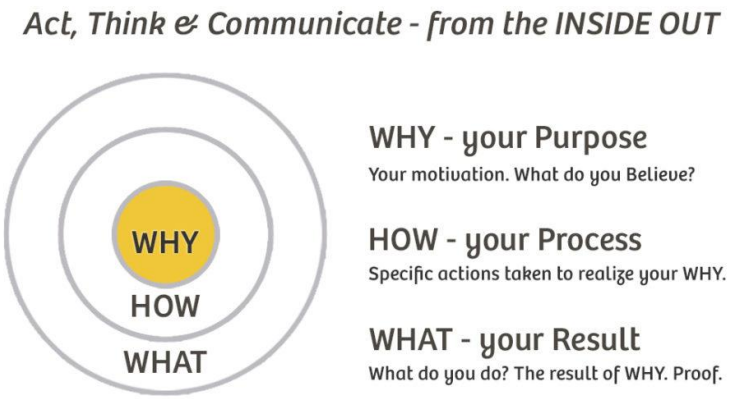


Figure 3 Golden circle

2.6 Sensation perception

It is critical in industrial design to understand the experience and perception of patrons. Anders Warell's Perceptual Product Experience model is a complete system by which to study these encounters. The PPE model describes three fundamental modes of experience: sensorial, cognitive, and affective.

The sensorial mode refers to the direct sensory stimulation a product elicits—what are users seeing, hearing, feeling, smelling, or tasting. For example, the texture of a material or the sound of a button click has the potential to hugely affect users' perception.

Cognitive mode goes beyond mere sensation. It encompasses the user's understanding and interpretation of that which they are sensing. It includes deciding why a product exists, why it works, and deciding whether to use it.

The affective mode concerns the emotions which a product induces. These may result from the product's appearance, meanings associated with the product or even memories it triggers. A good product may induce a feeling of trust, nostalgia or joy.

Together with Rocchi and Hammerman, Warell also identifies two fundamental aspects of experience: presentation and representation. Presentation refers to the uninterpreted experience as unveiled to the senses. Representation refers to the underlying meaning and connotation that users interpret through the application of personal memory, culture, and context.

The application of the PPE model in designing ensures a more comprehensive and sympathetic design. It ensures that the designer does not only concentrate on function but also on how people perceive products through their senses and emotions. It makes the design more fulfilling for users as well as more sustainable in terms of its relevance [17].



Figure 4 PPE model

2.6.1 Empathy map

Empathy mapping is a systematic framework used to externalize and conceptualize what is known about a user's internal and external world. It is an essential element of user-centeredness, especially in design and development work around human experience. The main purpose of empathy mapping is the creation of insight about users, most notably their perceptions, thoughts, feelings, and behavior, to make decisions.

The standard empathy map has four main quadrants: Says, Thinks, Does, and Feels. These are synonymous with various forms of user experience and help structure user data such that the nuances of human feeling and perception are articulated.
For example:

- Says is raw verbal feedback from the user, typically collected through interviews or user testing.
- Thinks gets into the user's internal thinking, detecting cognitive aspects that are not always verbalized.
- Does catches behavior and interactions with a product or environment that are observed.
- Feels maps out emotional responses, detecting affective states such as frustration, excitement, or confusion.

This model offers a rich set of lenses to look at how individuals experience and interpret stimuli, both at the raw sensory level (sensation) and higher levels of interpretation (perception). The zones of overlapping in the quadrants also reflect the richness of human experience, where cognitive, behavioral, and affective factors normally overlap.
Empathy maps tend to be created early in a project to collate research findings and identify areas of incomplete knowledge. If developed collaboratively, they can also be a vehicle for bringing teams together around shared user insight, removing individual bias and revealing underlying needs or pain points. In the end, empathy mapping creates a more nuanced, perception-sensitive design process that not only observes what people do but seeks to understand why they do it [18].

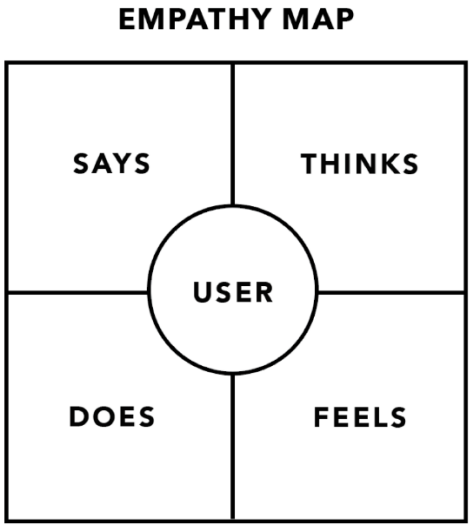


Figure 5 Emapthy map

2.6.2 Why we love and hate everyday things

The Juicy Salif citrus juicer, designed by Philippe Starck in 1990, is a widely recognized product that exemplifies the interaction between aesthetic appeal and functionality in product design. Its bold, sculptural form is designed to attract attention and provoke thought, making it an icon of contemporary design. The juicer, made of aluminum with a tripod-like structure, is visually striking, but its form is often criticized for being impractical in terms of usability.



Figure 6 Juicy Sarif designed by Phillipe Strack

This design choice highlights a critical point in product design: the relationship between form and function. While Juicy Salif's visual appeal is undeniable, its design sacrifices functionality. The juicer is notoriously unstable when in use, and its conical shape makes squeezing citrus difficult and messy. These design flaws underscore the importance of balancing aesthetic appeal with practicality.

However, Juicy Salif's success lies in its ability to generate an emotional response. It challenges traditional juicer design by turning an everyday object into a work of art. The sensory experience it offers is rooted in its visual impact, engaging users on an emotional level. This example highlights how emotional design can influence a product's reception. People may initially be drawn to its beauty and innovation, but the functionality of the product plays a crucial role in determining long-term user satisfaction.

In the context of sensory perception, the Juicy Salif serves as an excellent case for exploring how aesthetic appeal can evoke emotions, yet how the user's tactile and functional experience ultimately shapes their satisfaction. It emphasizes the need for designers to find a balance between aesthetic form and functional performance to create a truly successful product [19].

3 METHODS

3.1 Double diamond methodology

Double Diamond Method is used widely by organizations to incorporate design principles for the creation of innovative solutions as well as new ideas. For this project, the Double Diamond approach guides the entire product development process via four core phases: Discover, Define, Develop, and Deliver.

Discover: This is the initial stage that involves discovering the cause of the problem to grasp its purpose effectively. As a divergent stage, it tests out all the information and ideas via interviews, literature research, and focus groups to create a rich problem briefly.

Define: A stage of convergence wherein general ideas is formulated into a particular problem statement. This involves defining user personas, problem articulation, and storyboarding.

Develop: Another diverging phase, where various solutions are generated through ideation, rapid prototyping, storytelling, brainstorming, and concept development.

Deliver: A convergent phase that finishes the ideal solution into one product. This is accomplished through collecting user testing and expert analysis to look at the final design [20].

3.2 Discover

3.2.1 Market research

Market research is a systematic process of gathering data to make intelligent decisions. Its primary purpose is primarily to analyze firms in specified product or service categories to predict how consumers will react. It is one of the most effective ways of improving user satisfaction, customer loyalty, and profitability.

Market research in this thesis is employed to find out about trends and design themes in open-office spaces, examining how companies utilize these areas, their impact on productivity and collaboration, and the most significant design elements that optimize functionality. It is also employed to examine the market and identify gaps for desk solutions that cater to the needs of open-office spaces [21].

3.3 Define

3.3.1 Persona

Creating personas introduces designers to customers' needs, environments, actions, and goals. It is rooted in people's differences on an individual basis and instills empathy for people designers are going to design for. Personas belong to great user experience because they promote design conclusions that map to desired outcomes from users targeted. Using representative users, designers can effectively interpret research conclusions into impactful offerings for them [22].

3.3.2 Moodboard

It is a board filled with photographs, materials, and textures that evoke a definite mood, tone, or theme for a creative undertaking. A mood board uses a range of different forms of media, such as photographs, sketches, and textiles, all with one goal in mind: to represent that desired style or mood.

The mood board is mentioned extensively at an initial phase in design projects as an inspiration and reference source. The mood board draws attention to one project and is an excellent reference source in terms of how to handle a design project [23].

3.3.3 Requirement specification

The requirements specification process defines customers' expectation of how features, uses, performance, limitations, and reliability a product must possess. The requirements are mostly defined by detailing what must

be achieved or features they must possess. The customers also unconsciously communicate their needs. There also are underlying principles on how to react to these needs, and thus there is a possibility for the author to determine how to react to it. The design brief specification will have precise information on all elements that will assist in addressing customers' needs [24].

3.4 Develop

3.4.1 Sketching

One basic method that designers use to create, improve, perfect, and communicate their concepts is sketching. Sketches are low-cost, quick, and easy to correct or discard with little effort. In contrast to oral and written communication, sketching eschews formal constraints, making it an efficient communication method and an antidote to miscommunication. Sketching also allows constant idea generation and idea tests by means of additional sketches to ensure that the final preferred direction has adequately addressed the major concerns at stake [25].

3.4.2 Concept evaluation

Screening for most promising concepts is carried out on a two-stage basis: idea rating and concept evaluation. The idea is rated initially as rapid screening to select better options. The concepts are rated and compared in terms of pre-set criteria such that they can be compared by the team or author. The concepts are compared with a relative reference point with an evaluation grid. The process is repeated, and new concepts are formed by selecting other features from existing concepts [25].

3.4.3 Brainstorming

Brainstorming generates more ideas than are feasible when written down individually. Brainstorming allows for collaboration with participants contributing and building on other participants' thoughts to maximize opportunities for creativity. Brainstorming allows designers and researchers to sharpen their creative thinking skills by suspending analytical thinking temporarily. The method can effectively be used at all phases involved in product development to maximize opportunities for exploration and come up with more innovative and successful outcomes [26].

3.4.4 Mockup

Mockup is a critical design development aid used by designers, engineers, and stakeholders to envision, examine, and perfect a product's design and functionality before progressing to later stages. Good mockups save businesses money by addressing development issues early on, streamlining development, and making projects more successful [27].

3.4.5 CAD

Computer-aided design (CAD) is software used to convey concepts as a group of three-dimensional physical objects, typically created from basic geometric shapes. One of the main advantages of CAD is that it allows designers to visualize the designs in 3D. CAD designs can also be used as trial mockups, which in some instances can replace physical models. CAD designs are sometimes utilized to painstakingly craft the entire construction of the final product and analyze how the components interact [25].

3.4.6 Rendering

Product rendering is also known as computer rendering or 3D rendering. It is a process utilized to obtain photorealistic product images out of product designs created in production and design stages. Via 3D rendering tools, designers can create virtual product images without them ever having existed physically, making them able to analyze and comment on them instantly at no prototype cost. It is an asset not only to utilize for presentation in design but also for visualization, as it smoothly reflects a product's shape, materials, and finishes. Besides, rendering also allows us to create various variations of a design, e.g., other materials and colors, without having to create several physical models. It also provides designers with allowances for placing them in other

environments with complete creative freedom. Product rendering is generally an affordable and faster way by which product designs are displayed, making it an integral tool in marketing and design stages [27].

3.5 Deliver

3.5.1 Prototype

This is the process of developing a model of what the final product or idea will appear like. Physical mockups are copies that most closely resemble the final product as far as how it looks and feels. Concept mockups are employed to quickly evaluate an idea and try previously untested parts of a product to determine if it will work. Prototypes offer interaction among different stakeholders such as designers, project collaborators, team members, and clients. This is especially true for physical models, since an attractive, tangible 3D version of a product is far more comprehensible than a written description or sketch. The concept models serve to ensure that all the parts and components of a product are functioning as they should [25].

4 Implementation & Approaches

4.1 Discover

4.1.1 Market research

The market research stage gave one an idea of the trends in the sizes, shapes, and wood type for furniture products, especially lounge and coffee tables. The study enabled one to have a clear picture of the prevailing tastes in aesthetics, the expectations of customers, and the shopping behavior in terms of the design of lounge tables.

A survey of the market for products indicated that tables have very differing dimensions depending on their utilization. Coffee tables usually range from 40 to 50 cm in height as they are intended for personal spaces like living rooms. They are functional in that they provide a space for snacks, decorations, or miscellaneous items in rooms that have sofas and TVs.

On the other hand, lounge tables are lower, ranging from 30 to 40 cm in height. This reduced height is suited for social settings, where the table should not serve as a barrier to communication or subtract from eye-to-eye contact among individuals seated on adjacent furniture. These ergonomic and spatial considerations come heavily into play in the design and sizing of lounge tables in contemporary market offerings.

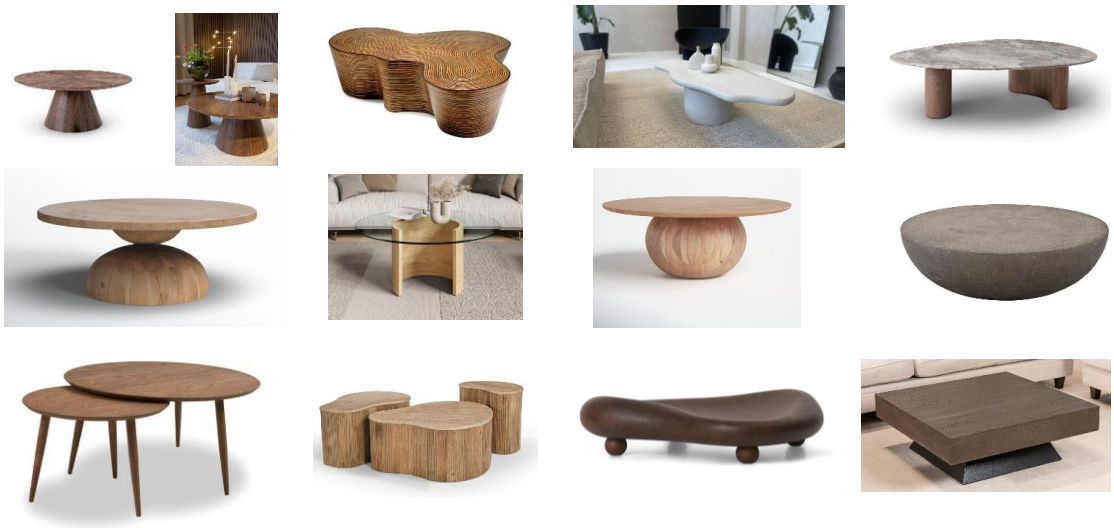


Figure 7 Pictures of lounge table from the market research

4.2 Define

4.2.1 Personas

For the ease of determining the target market of the product, two main personas were established. The differentiation needs to be made since office spaces are usually designed by professional interior designers who are hired to organize the spatial and aesthetic layout. Interior designers undertake their work with a keen design philosophy, appreciating the values intrinsic in every material and empathizing with the general environment. Designing an effective workspace is crucial since there have been many studies proving that a well-designed environment greatly improves well-being and productivity.

Though the interior designer has a large role in determining the appearance and function of the room, the ultimate purchasing decision generally lies in the hands of the office manager. This is the person who hires an interior designer and makes sure the furniture chosen reflects the values of the company. To the office

manager, it is significant that the furniture is an ethical and sustainable option, i.e., the utilization of green materials, human rights respect, and workplace health concern. The selections then become an outward expression of the company's values and identity.

Olivia Andersson – Sustainable Lounge Designer



Name: Olivia Andersson
Age: 38
Location: Stockholm, Sweden
Education: Master's in interior architecture
Experience: 12+ years in commercial interior design

About Olivia

Olivia is an interior architect specializing in lounge spaces for offices, hotels, and corporate environments. Passionate about nature and sustainability, she designs inviting, eco-conscious spaces that reflect a company's brand while prioritizing comfort and functionality.

Design Philosophy

Nature-Inspired: Incorporates natural materials, greenery, and organic shapes to create calming environments.

Sustainable Focus: Uses reclaimed wood, energy-efficient lighting, and eco-friendly fabrics

Clients & Work

She collaborates with hotels, corporate offices, and co-working spaces, crafting elegant yet sustainable lounges that leave a lasting impression.

Figure 8 Persona 1 interior architect

Erik Lundström – Office Manager & Lounge Project Lead



Name: Erik Lundström
Age: 42
Location: Stockholm, Sweden
Role: Office Manager & Workplace Strategist
Company: Mid-sized tech firm with a growing team
Experience: 10+ years in office management and employee experience

About Erik

Erik is responsible for creating a productive and inviting office environment. With his company expanding, he sees the need for a well-designed lounge area that fosters collaboration, relaxation, and a strong company culture. He values aesthetics but prioritizes functionality and employee well-being.

Goals & Vision

- Create a lounge that enhances productivity and team interaction
- Ensure the space reflects the company's brand and values
- Use sustainable and high-quality materials for a long-lasting design

Figure 9 Persona 2 office manager

4.2.2 Moodboard

The goal of a moodboard is to capture the emotional tone and atmosphere that the design concept will evoke and provide a shared visual narrative throughout the project. It is used as a directional tool, and specific images and words are meant to motivate and remind one of the intended feelings and direction of design.

A central theme in the moodboard is the connection between organic forms and nature. Many shapes found in nature are inherently organic, and this visual language aligns closely with principles of biophilic design. Biophilic design has been shown in various studies to enhance well-being and productivity in workplace environments by fostering a closer connection to nature.

The moodboard also conveys the brand values of Sense by Us, which is positioned on creating exclusive, customized furniture solutions. The theme of exclusivity is defined by limited availability, either to one person or an elite group, and is conveyed through custom design rather than mass production. Exclusivity may also be expressed through personalized user experiences, premium materials, and controlled distribution.

Minimalism is another dominant concept conveyed in the moodboard. The design should be integrated into its environment without overpowering it. Minimalism creates visual consistency and reduces cognitive load, which makes the environment feel calm and uncluttered. It can be achieved through design language consistency, timelessness, simplification of forms, and the intentional elimination of distractions.

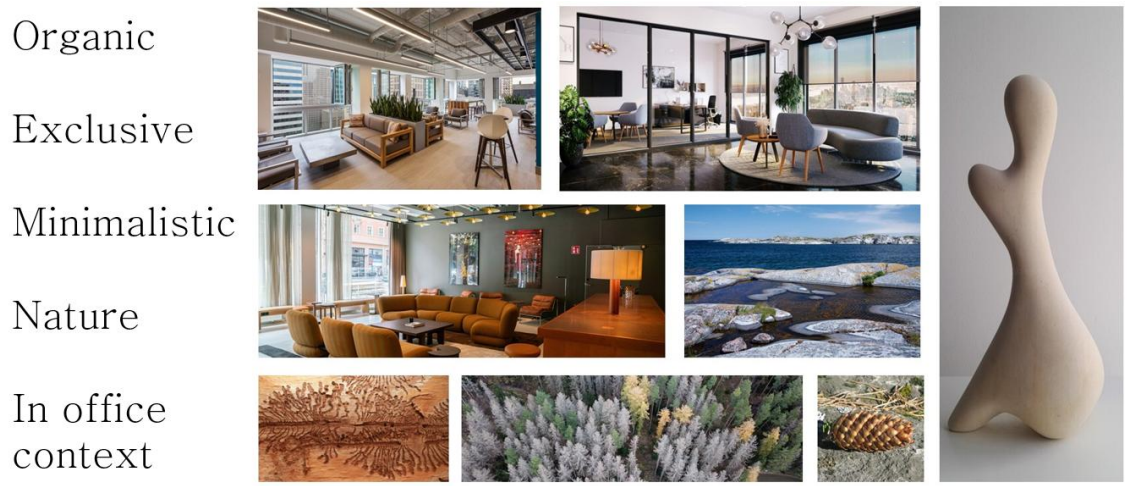


Figure 10 Moodboard and keywords

4.2.3 Requirements specification

The lounge table product features depend on the mission and values of the company as well as on what the target users and buyers require. The product needs to include the company's commitment to exclusivity, sustainability, and aesthetics in conjunction with functionality and suitability for modern-day workplace environments.

The major requirements for the lounge table are:

- Brand value alignment: The product should reflect the personality of the brand by showcasing an innovative, sophisticated product that is premium in nature and thoughtful.

- Sustainable use of materials: The materials used should be environmentally compatible. The table should have less impact on the environment at extraction and production stages. In this item, spruce wood affected by spruce bark beetles were used, which ensures the utilization of otherwise wasted natural materials.

- Subtle but unique design: The table should be plain, refined in design to fit seamlessly into modern lounge or office environments, without appearing as though it may also pass as a duplicate product already available on the shelves.

- Standard size: The table is designed to meet functional and ergonomic needs by adhering to standard measurements for lounge tables. It has a 40 cm maximum height to enable its use in social settings.

- Manufacturability: The item should be manufacturable with available tools, methods, and an acceptable cost and timeline constraint without sacrificing its quality.

- Structural stability: The table should have structural stability and should easily support day-to-day objects such as books, laptops, drinks, or decorations.

- User safety and comfort: The edges and corners should be curved and smooth to avoid hurting anyone. The table should also be heavy enough not to topple over but still lightweight enough to move.

These standards form a basis for development, and therefore the final product is not only functional, but also a great representation of company values and design principles.

4.3 Develop

4.3.1 First iteration Sketches

At the outset of the design process, Sense by Us had no preconceived notion of what the final appearance of the product would be. To initiate dialogue and come to an understanding of potential directions for design, a series of exploratory sketches were developed. These initial sketches did not try to define precise shape or form but were meant to explore various functional potentials and intents of design.

The focus was placed on investigating key issues such as whether the table should include adjustable height mechanisms, integral storage elements, a material or visual connection to nature, or the incorporation of pattern and texture. By presenting an array of conceptual ideas, the sketches were utilized as a tool for discussion, allowing the company to establish likes and values that would be able to drive the further development of the lounge table design.

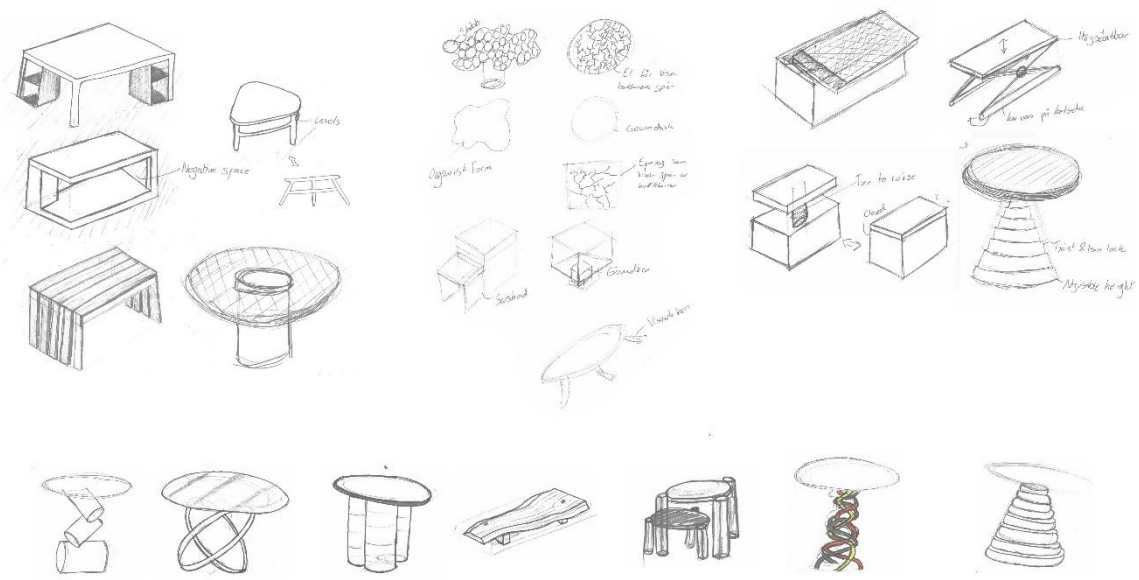


Figure 11 first iteration sketches

4.3.1.1 Second iteration sketches

Sense by us made it clear that they want the product to have a connection to nature, and the idea about connecting it to the beetle made it look match with the moodboard and was something new to the market, with a clear story that can be used as a selling point for the company, there was some different ideas of how the base of the table could look like.

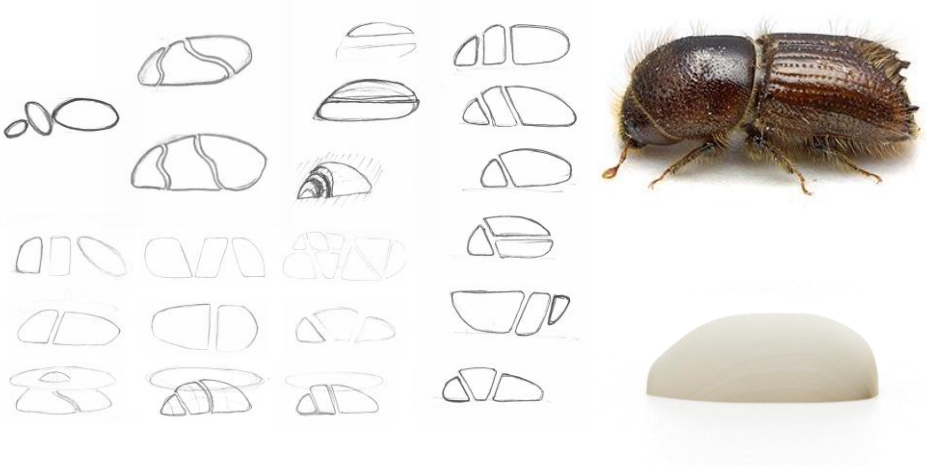


Figure 12 second iteration sketches

4.3.2 CAD

These design concepts were developed with the assistance of CAD (Computer-Aided Design) software to assist in an optimal and accurate design process. As these models were created digitally, there was also the possibility to iteratively improve them quickly, build mock-ups, develop photo-realistic images, and get the designs ready for production in-house.

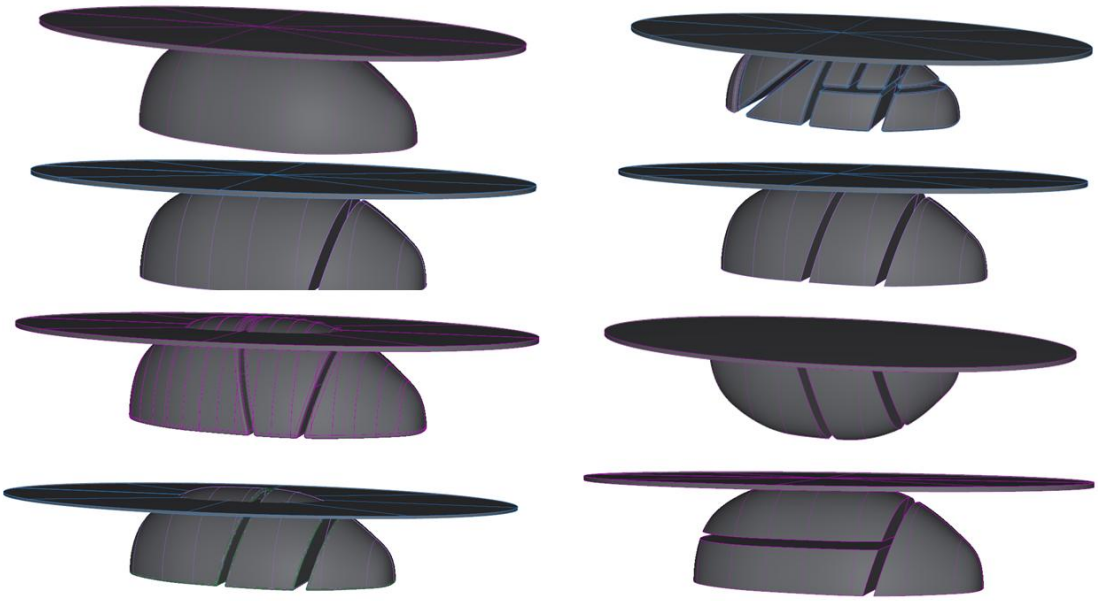


Figure 13 CAD

4.3.3 Mock-up

For greater visualization and examination of the idea in a physical manner, a series of quick mock-ups were produced with 3D printing. The models were pivotal in fully understanding the size and proportion of the lounge table and gave a physical reference point that enabled spanning digital CAD models, projections, and real-life perspective.

By having physical models, it was easier to test how the design would function in space, how the shape would appear from different angles, and whether the size would be suitable for a lounge table in a social environment.

Mock-ups enabled iterative design through feedback cycles, with the opportunity to make alterations before actual prototyping. This hands-on method saved resources and time, as well as keeping the final idea on track with intended user and brand values.

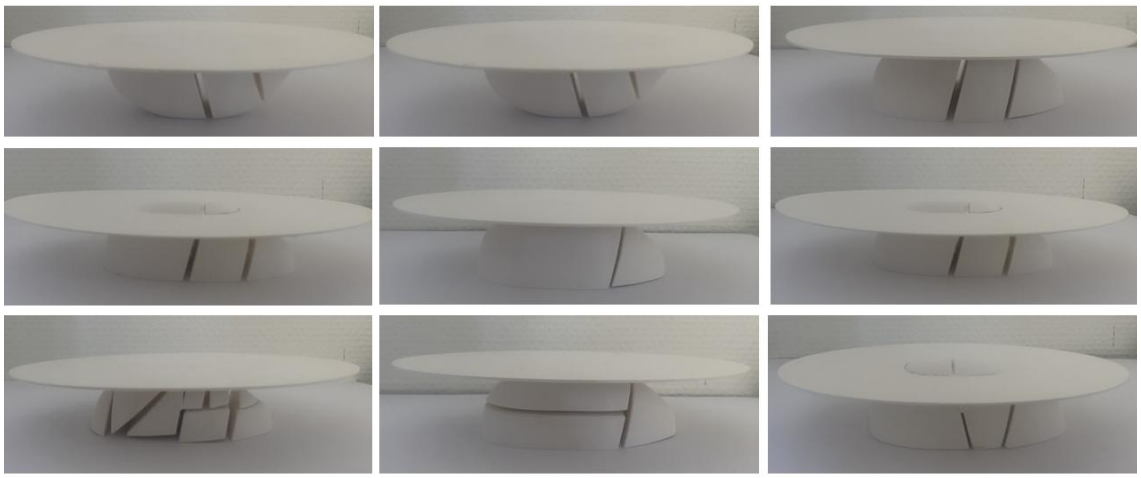


Figure 14 3d printed mockups

4.3.4 Rendering

To provide a realistic portrayal of what the final design would appear like with material and lighting added, KeyShot Studios was used to create high-quality renderings of the lounge table. Renderings played a vital role in communicating the design intent, especially in terms of materiality, surface finishes, and how the product would interact with its environment.

KeyShot allowed for the precision application of selected materials, such as beetle-damaged spruce wood, while allowing the freedom to experiment with different combinations of materials, finishes, and color schemes. This allowed for a determination of how different selections would affect the overall appearance, mood, and harmony of the piece.

By generating realistic images, the renderings helped to direct internal decision-making, facilitated company discussions, and served as an excellent presentation tool for stakeholders. They also provided an early glimpse into how the lounge table would integrate within several interior settings, aligning with the aspirations of minimalism, exclusivity, and sustainability.

They created an essential connection between concept and actuality, ensuring that the final product would not only meet functional and material requirements, but also incorporate the emotional and aesthetic qualities desired from the beginning of the design process.



Figure 15 Renderings of some of the second iteration sketches

4.3.5 Concept evaluation

Through the initial set of sketches and discussions, Sense by Us favored a product story that was connected to nature. They emphasized the importance of storytelling through design, especially one that could capture sustainability and environmental awareness. The idea of making use of beetle-killed spruce wood not only appealed to these ideals but also offered a robust and authentic story behind the product.

The initial ideas' abstractness was appreciated; the company favored the organic and artistic approach. However, the criticism noted that some ideas were too abstract and could make it harder for the message and the narrative to penetrate. Henceforth, they shifted towards an idea that was balanced, something visually unique and straightforward but more perceivable with reference to the beetle and its impact upon the wood.

The popular concept incorporated a subtle yet purposeful reference to the natural devastation caused by the beetle, turning what is otherwise considered a flaw into something celebratory as a design feature. This not only reinforced the brand's focus on sustainability and scarcity but also infused emotional and environmental depth into the final product.

The process of evaluation was key in getting the design balanced to the company's values and selecting a path that would resonate with both designers and consumers. It assisted in ensuring the ultimate concept for the lounge table was not only aesthetically and functionally strong, but also meaningful on a story and material choice level.

5 Result

5.1 Result of research

5.1.1 Sustainability

A very important decision during the development process was material choice. It was clear from the start that one of the materials would need to be spruce wood infected by the spruce bark beetle since this has a direct relation to the product's conceptual and environmental message. What was not so clear, however, was whether the product would be made exclusively from this wood or if it would be combined with other materials.

As the design progressed, particularly the idea of shaping the table base into an abstracted form of the spruce bark beetle, it became clear that a wood base paired with a glass top gave the most compelling aesthetic and conceptual solution. Alternative pairings of materials were tried, such as an aluminum base with a wood top, but these failed to deliver the uniqueness and visual cohesion desired in the completed product. Since sustainability lies at the heart of the design's purpose, wood as the primary material for the base felt more organic and intentional.

The use of glass for the tabletop was driven by practical as well as symbolic considerations. A transparent top will maintain the visibility of the base's shape from all angles, especially important due to the table's low height, which would otherwise obscure the base's intricate shape. While other transparent materials such as acrylic or epoxy were considered, glass was ultimately chosen for its premium feel, longevity, and fit with the sustainable and premium positioning of the product.

Production will take place in-house at Sense by Us' workshop, allowing for full control over the production process and high-quality craftsmanship. All materials are sourced from local suppliers, supporting local businesses as well as reducing transportation emissions significantly. The local approach adds to the product's overall environmental footprint and is a fundamental element of the project's commitment to sustainability.

5.1.2 Desirability

For desirability, therefore, the lounge table needed to be attractive as applicable and functional to its target clientele of designers and office managers. It needed to appeal also to its environment for use. The product will serve in hospitality environments and in office lounges, where understatement is key. It fits easily into an array of interiors with its neutral and nature-inspired theme.

Material and form choices were critical to making this possible. Use of wood damaged by spruce bark beetle added an emotive element, literally connecting it to environmentally conscious practice and imparting to it a visual uniqueness. It was topped with glass to make its sculptural wood base fully viewable from all its angles, marrying functionality to visual narrative. Its transparency also has a role in contributing to cleanliness and simplicity without sacrificing clarity in its structural composition.

This reflective design process was characterized by constant experimentation and reformulation. For instance, initial experimentation with materials investigated potential utilization of an aluminum base. However, through iterative prototyping and sketching, it became evident that aluminum was not able to convey wood's tactile warmth or its emotive qualities. This reflective process of consideration and adaptation at play resulted in the conclusion to utilize a solid wood base, both narratively as well as aesthetically.

5.1.3 Feasibility

The function of the lounge table is not only to bring aesthetics and high-quality value to the rooms it holds space in but also to signify the core values of the organizations that choose to incorporate it into their use. While qualities such as elegance, individuality, and longevity regarding product lifespan matter, none of these are worth anything if the product itself can be created.

To enable the production of the product successfully, the base was divided into 40 mm thick sections. This was in terms of having timber available and the convenience of handling thin cuts. Thick wood is less accessible and more expensive due to the limitation in supply. Each base section is machined by a CNC router to make with precision the half-circle shapes which together form the sculptural base of the table.

Assembly alignment is facilitated by the inclusion of four dowel holes per segment as guides for proper positioning. Small holes were also added to indicate orientation and stacking order of the pieces, thus facilitating correct assembly.

With the segments milled and aligned, the specialty cuts unique to the design according to the body structure of the bark beetle are made using a bandsaw and chainsaw setup. While these are specialty tools involving some risk when used separately, Sense by Us woodworkers possess considerable experience and have worked on such projects before with success.

Finally, the glass surface is secured onto the wooden base with four threaded rods, being stable without disrupting the minimalist aesthetic. Stability is achieved in this type of fastening while the clean, non-intrusive appearance of the table is not compromised.

5.1.4 Viability

The product's narrative emerged from the selection of beetle-damaged spruce as the primary material. This wood, which is usually discarded or used for firewood, had been intentionally chosen to highlight a pressing environmental issue that Swedish forests are experiencing.

All the materials were local, and the product is made in-house at Sense by Us. This allows complete control over workmanship and minimizes transport-based emissions. All these choices not only support local industry but minimize the environmental impact of production.

The use of reclaimed material is also aligned with the company philosophy and values of sustainable, narrative design. While this narrative will be explored more fully under Discussion (6.1.3), it was cultivated as a component of final product form.

5.2 Result of design proposal

5.2.1 Shape and Expression

Sense by Us' design group was immediately attracted to the spruce bark beetle form. It was chosen because it had great symbolic appeal to nature and to Swedish forest owners' immediate issues. The abstract nature of the form, where its actual meaning isn't revealed until its tale is shared, was an aspect that they found particularly intriguing. They appreciated how this ambiguity promoted curiosity and was solidly rooted in a proper story.

What made the design's biggest achievement was the balance it achieved between natural inspiration and a clean, minimalist, and elitist appearance. This harmony of form and concept helped to give the team members an intense feeling of satisfaction with the outcome.

The form of the product itself is also directly taken from the bodily shape of the spruce bark beetle, which is typically split into three distinct parts. To replicate this, two inconspicuous cuts were incorporated into the product, in the same alignment and proportional size ratio as in the body of the beetle. These cuts are both a visual signal and a design element that connects the product to its biological model in an important and elegant way.

5.2.2 Tabletop

During the development phase of the design, there were many discussions about what the perfect shape of the glass tabletop was to be. Geometric and organic shapes were both considered, and they were created and put into consideration along with the wooden base to try and find the best matching pair.

By following this back-and-forth design process, it became evident that the oval shape was most aesthetically and conceptually suitable. An organically shaped top was tried to represent the organic impact of the wooden base, but they didn't have the consideration and visual nuance desired for the overall design. Conversely, the oval provided balance and poise that would complement the sculptural beetle-shaped wooden base without overpowering it.

The choice to merge the natural textural wood base with the smooth oval-shaped glass top was one made by both the working company and design team. Geometric simplicity of the oval top provided counter to the expressive irregularity of the base, and they were thus able to achieve the goal of having both functional design and balance.

Other top designs from the initial development stage were set aside due to the fact they were failing to live up to the client's expectations. The oval top, however, was identified as the most appropriate solution, bridging the organic nature of the material to the visually oriented structured finish, which facilitates functional as well as design harmony.

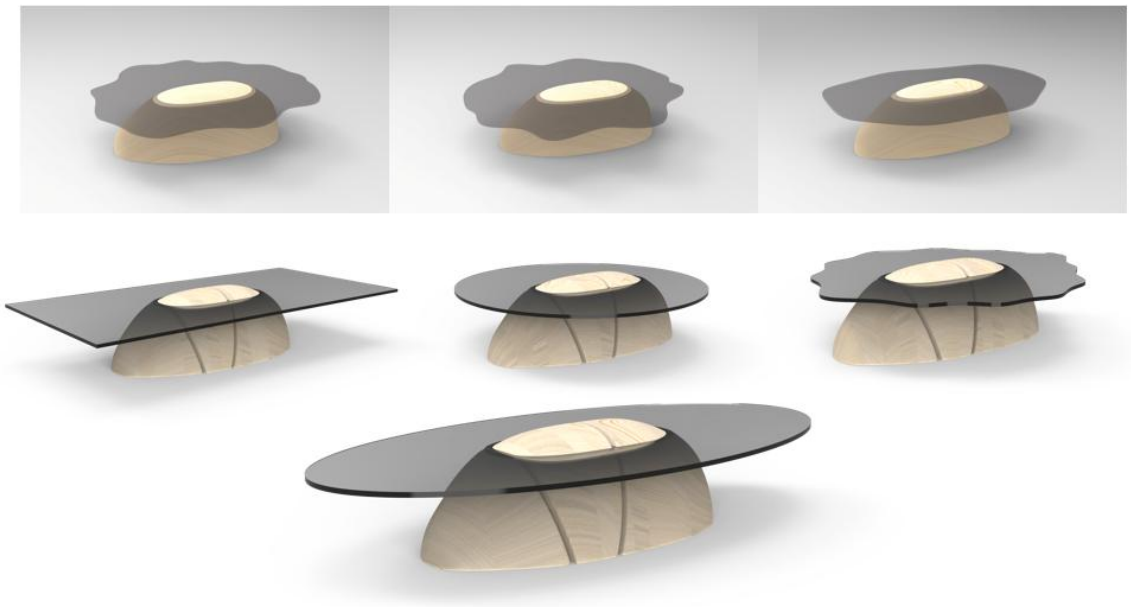


Figure 16 Renderings of different tabletops

5.2.3 Attachment between tabletop and base

The table's attachment to the base is both aesthetically and structurally vital to the finished product. The base in this design is created by semi-circular hollow cutouts. This reduces overall material use, encourages sustainability-based design, and reduces overall weight of product to aid in easier shipment and handling without sacrificing strength.

It has four threaded rods with an M10 diameter to give a strong and stable connection of components. The rods are threaded vertically through the inner framework and fixed by hidden nuts underneath the base. In this manner, there is a strong mechanical connection with no visible fixing components, ensuring visual purity of design.

Concealing how it is attached also relates to product expression, where construction is centered on function without overpowering form. By not showing mechanical elements, having a slim, clean design is possible for the product. The users' sense of quality is reinforced by minimalism and gives an impression of craftsmanship and thoughtfulness.

Thus, not only is the attachment functional, but it also serves to reinforce, unobtrusively, the overall design idiom of the product, such that form and function may be maximized by the user without interruption.

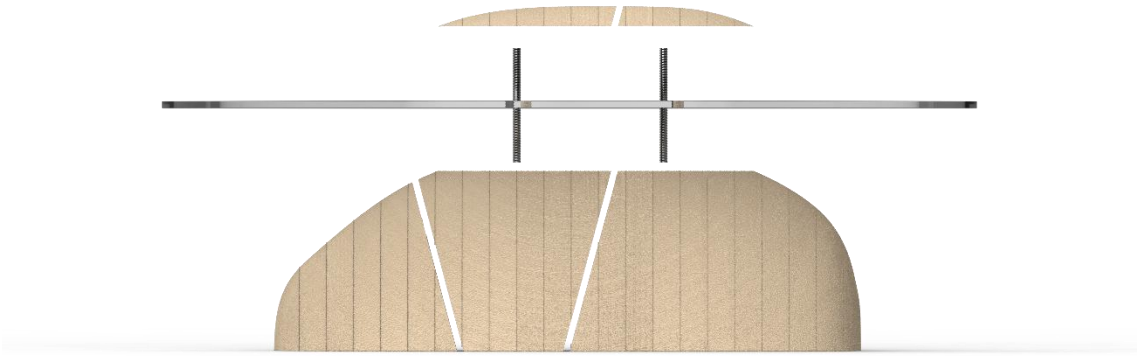


Figure 17 Exploded view of concept

5.2.4 CAD

In the CAD file, the model was prepared for production. This included splitting the base into 40 mm thick layers, since this is equal to the available stock sizes of the company's wood material. Each layer was also modeled as an arc to create a hollow internal structure, which reduces material usage and overall weight while maintaining stability.

To facilitate simple and accurate assembly in production, each layer has pre-drilled dowel holes. These allow wooden dowels to be inserted through and to act as guide rails so that each block will become correctly aligned when being glued together. Additionally, small reference holes were added on each segment for the further simplification of the placing and orientation, leading to simple and quick construction.

Utilizing CAD at this phase not only enables accuracy and control over the design but also serves as a bridge between digital planning and physical production. It simplifies communication with the production partners and minimizes the potential for errors during assembly by building clarity and logic directly into the design files.

5.2.5 E-drawing

E-drawing was also used to CNC-cut all pieces directly out of wood stock for precise fabrication and eventual assembly into the final product. This format is critical in ensuring that production is carried out precisely and as intended.

By providing finished electronic files with accurate dimensions, E-drawings help ensure that every part is cut to a high degree of accuracy. Not only does this ensure that every part will fit together as intended when assembled but also increases the likelihood that the final prototype resembles its original design idea.

Further, E-drawings also act as an excellent communication channel connecting production and design phases. E-drawings minimize miscommunication and production errors by creating a clear, standardized reference point for both workshop personnel and machine users. This way, E-drawing utilization directly improves prototyping process reliability, productivity, and quality.

5.2.6 Render

To better envision the resultant design with maximum realism, the high-detail renderings in 3D were produced using KeyShot. For pre-production viewing and communication, the renderings were extremely useful. With

the true-to-life light conditions, material, and background setting, the renderings provided the perfect fore view of what the lounge table will look like under real-life conditions.

The images obtained portrayed the unique nature of the spruce wood and glass top, which had been beetle damaged. This allowed for further investigation of the visual relationship between materials and where the table might sit in various interior environments. They were particularly helpful for internal discussion and relaying the idea to the partner firm.

Furthermore, the renderings were a visual guide to support prototype development and were still being used to try out different configurations' appearance without producing multiple iterations. In the process, the renderings assisted not only in visual storytelling but also in design verification and attaining sustainability, minimalism, and emotional connection goals of the endeavor.



Figure 18 product rendering

5.2.7 Contextual render

To better examine how the design will fit within its intended setting, context renderings were created to view the lounge table in real-world interior settings. The renderings placed the product in decorated lounge and office settings to assess how the product interacts with surrounding furniture, lighting, and room layouts.

Contextual photos provided us with insight into the proportions, size, and overall visual impact of the table positioned in different rooms with furniture. They demonstrated how the organic curve of the wooden base is paired with the linear shape of the glass top to provide complementary harmony in both contemporary and nature-oriented interiors.

These depictions were a big part of the communication of the design to the stakeholders and potential users. They were used to convey how the table is integrated into the overall aesthetic and atmosphere of the space, advancing its theme of responsible design, luxury, and sustainability.

By sitting the table within its potential uses, the context presentations bridged the conceptual/reality gap. They showed not only how the product could function as a table within the context of office space in general, but how it might have its part to play in the emotional and visual impact of the space.



Figure 19 contextual rendering

5.2.8 Prototype

A prototype lounge table at full scale was manufactured in Sense By Us' workshop. Production commenced with preparation and selection of spruce boards. The boards were clamped together to form wider boards to use in creating the base. The planks were then cut into equal dimensions after they had been bonded.

To achieve consistency across the boards, thicknessing and planning was done. The next step was surface refinement to evaluate the grain structure and search for any form of defects that would weaken finish or structural integrity.

With a ready supply of material, the design was arranged in a diagonal pattern utilizing CNC-routed templates. Paper templates were utilized for aligning templates to the correct dimensions of each tabletop. CNC cutting was done to produce segments one by one, one each for one layer in each part for the overall shape for a lounge table. These were properly arranged and made ready for assembly.



Figure 20 Initial stages of the prototype production. This includes lamination of spruce planks, preparation of raw material, CNC milling of component templates, and organization of the cut segments that will be layered into the final table base.

These base curve pieces were manufactured by means of CNC shaping. The pieces were cut in sections and then marked and aligned to ensure accuracy in stacking. The pieces were stack-built step by step to get to the full curvature of the base. Trimming and sanding were also done to prepare each piece before final assembly.

These were then glued and clamped to keep the structure in shape. Clamping systems were also utilized to keep pressure upon the compound curvature while curing. A jig was also made to hold the structure restrained in a fixed position and aligned while sanding and shaping.

The curved, organically shaped body of the lounge table was created by mechanical sanding devices. This required a lot of patience and inspection to get smooth, even finish without sacrificing the intended form of the design. The procedure concluded with a fully shaped base with a finish ready for final surface treatment and tabletop attaching.



Figure 21 Forming and shaping the table base. CNC-cut layers are glued and clamped into the intended shape, then fixed into jigs for alignment and stability. Rough shaping is followed by hand sanding to achieve the organic curvature of the lounge table's

After the entire base scale was constructed and sanded, the second step was cutting the base halfway through for the integration of structural connectors and ease in mounting the tabletop. Cutting halfway facilitated precision on the fitting of the glass panel into shape as required and the stability of the table.

After the halving of the base, further fine adjustment was done on the surface with the aim of removing imperfections and achieving level curves. Visual checks and hand adjustments ensured the two base halves would be perfect when joined together.

The attention was subsequently shifted towards the top surface of glass. An oval-CNC acrylic glass had precise holes drilled into it for it to be fixed onto the base with the use of concealed threaded nuts and rods. The fixing points were recessed in the wooden frame such that the fixings were not visible from the outside.

After the two base parts were ready and the tabletop was also as desired, the two were assembled at last. The two base parts were joined and fixed with the threaded rods, and the glass display top was set and fixed in place, finalizing the prototype assembly.
The complete lounge table was placed in an exhibition environment, conveying the final design in the finished form. The exhibition highlighted the material origin of the table, beetle killed spruce, as well as the expressive, clean shape and sustainable design. The finished product was a functional and visual representation of the entire design process, ranging from the idea, the production, up to the exhibition.



Figure 22 Sawing and assembly. The prototype is cut and refined using a bandsaw for precision. The base is shaped, cleaned, and joined with the top support element. The tabletop holes are aligned, and the glass top is prepared for mounting.

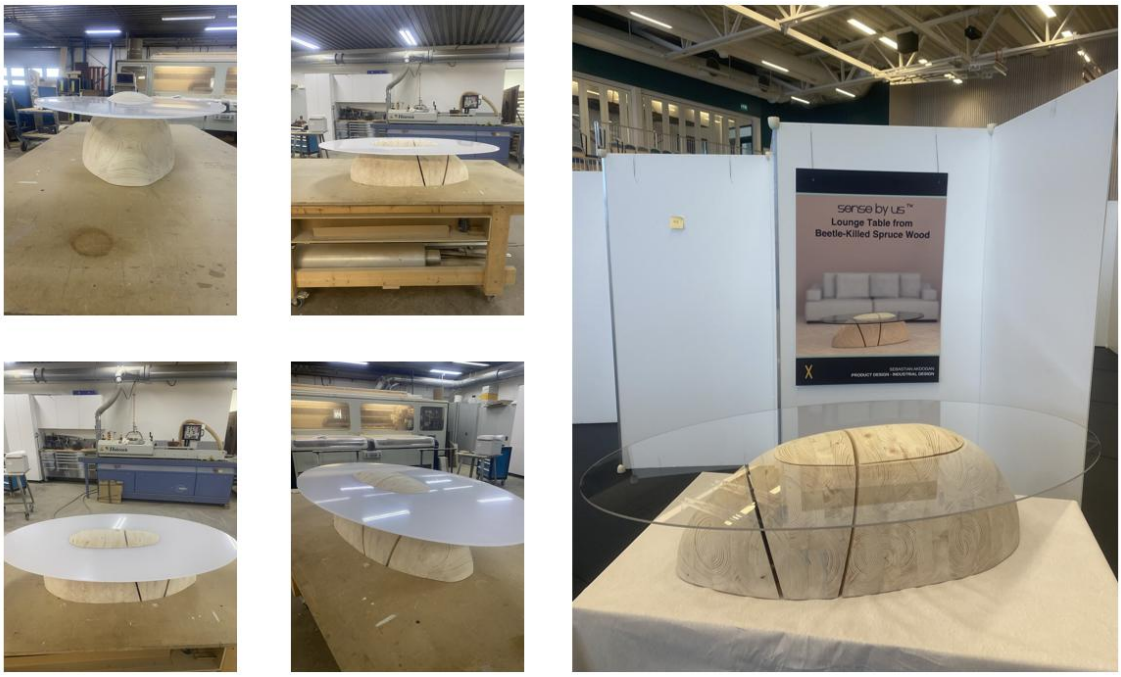


Figure 23 Final assembly and exhibition. The glass tabletop is fitted to the finished wooden base. The complete prototype is displayed in a showroom setting with supporting presentation materials, showcasing the final design outcome of the thesis project.

6 Discussion and conclusions

6.1 Discussion on research questions

6.1.1 How to design a meaningful lounge table considering desirability, viability, sustainability and feasibility?

Designing a symbolic lounge table requires an integrated strategy finding a balance among aesthetics, functionality, and storytelling with environmental and production requirements. This project investigated how the principles of desirability, viability, sustainability, and feasibility can be read as functional design decisions to create a product that speaks to the users as much as the producers.

For desirability, the lounge table was conceived with a clear user and context in mind. Modern living room environments such as office spaces, hotels, and interior spaces in public places affected the design approach. The form language is subtle, neutral, and minimalist to allow it to blend with various environments without diminishing its effect. The emotional appeal that is generated by finding inspiration from the spruce bark beetle in an abstracted sculptural form contributes to the emotional appeal of the product. It has a hook of intrigue and storytelling that makes it more than average as a regular piece of furniture.

Viability was addressed through narrative and brand congruence. The product narrates a powerful story attached to a relevant ecological cause: the damage done by the spruce bark beetle to Swedish forests. Made from locally harvested, beetle-killed spruce, the table is a means of environmental awareness. Not only does this enhance the product's uniqueness, but it also provides businesses with a means of communicating their values through interior choices. Their investment is a statement of environmental and social responsibility.

While we believe that this product has strong potential to be a success because of its history, sustainability, and appearance, we also must acknowledge that it is only when it is out in the market and embraced by consumers that we can finally justify it. Until then when it is bought and used, its final viability remains untested.

Sustainability guided several design decisions, including material, production location, and modularity of the base. Local wood and local production cut carbon use and support local enterprise. Glass was selected for the top because it is durable, recyclable, and can highlight the wooden base beneath. The combination of these materials results in long-term use and sustainability.

Lastly, feasibility was ensured by keeping the base as modular components, and compatible with existing CNC technology as well as available stock dimensions (40 mm boards). The production process was simplified to avoid waste and complexity, while allowing low-volume production without compromising on quality or stability. The assembly process is done with dowels, pre-drilled guide holes, and threaded rods. It was made repeatable, secure, as well as non-obtrusive visually.

In addition to the implementation of these principles, the design process was one that adhered to Donald Schön's reflection-in-action. Rather than working from a dogmatic approach, the process was that of iterative, experiential conversation between material behavior and design intention. As an example, early designs with aluminum on the base were dismissed later after mock-ups and sketches revealed an incompatibility of harmony with the theme of sustainability. This turn to reflection drew into sharp relief the value of working with the tactile, local wood, amplifying the emotional and environmental worth of the product. These reframing moments, guided by present-time insight, demonstrate how reflective practice enriches both process and product.

6.1.2 What is a meaningful product?

A great product is not just functional. It is full of intention, meaning, and emotion, and communicates with its users on a profound level. It is infused with cultural, environmental, and social values and still performs a functional role. In today's design landscape, where users expect authenticity and value alignment at an individual or organizational level, meaning has become the ultimate metric for product success.

In the context of this project, the lounge table is given meaning by its strong narrative connection to an environmental cause: the widespread destruction caused by the spruce bark beetle in Swedish forests. Through recycling damaged wood that might otherwise be discarded or burned, the product gives new life to a material that signifies loss and reconfigures it as a statement of resilience and rebirth. This makes the product from a simple object to a talking point, one that is conscious of sustainability and the situation of forest owners.

Besides, the meaning lies in the form, material, and method of manufacture. The abstract but recognizable form, taken from the broken-up body of the beetle, adds a poetic element to the design. The users won't necessarily be aware of it, but once they recognize the reference, their emotional connection with the product is added to it. Not only does the glass tabletop provide transparency in a physical sense, but metaphorically speaking, it also symbolizes openness so that the story of the base always remains visible.

The product is also important in terms of how it aligns with existing values of local production, sustainability, and aesthetic restraint. It is respectful of the environment, supports local industry, and is constructed to last stylistically and physically. These considerations appeal not only to individual users, but also to firms and institutions who wish to communicate their values through their environments.

In effect, a meaningful product is one that people can relate to on more than one level. It does something, it has a great story to tell, it helps solve real-world problems, and it makes one feel proud or responsible to own. This project illustrates that if purpose is ingrained in each design choice, it's achievable to make a product that is beyond utility and into the emotional and cultural realm.

6.1.3 The role of storytelling

The lounge table is not merely an item but carries with it a narrative intended to appeal to both consumers and users. The product tells the story of an environmental issue that occurs in Swedish forests, that being the destruction caused by the spruce bark beetle. By repurposing the spruce wood that would otherwise go to waste or be incinerated, the table gives new life to material representing environmental loss. This transforms the product into a symbol of sustainability and resilience.

Storytelling here is commercial and emotional. It allows users and companies to subscribe to values of craftsmanship, sustainability, and local production. Companies that furnish their offices with these products make a public statement of their environmental and ethical concerns. The table is a conversation piece that provides the space with identity and depth.

From a commercial perspective, this story brings in differentiation and emotional engagement, making the product more sustainable. Emotional connection is increasingly driving design-led buying decisions and meaning-transmitting products are more likely to be remembered and respected over the longer term.

6.1.4 Contribution to the field

This thesis contributes to the field of industrial design through a theoretical and practical example of the combination of sustainability, narrative and emotional design in the field of furniture design. Using beetle-killed spruce wood as a case study, the thesis can potentially turn rejected or devalued materials into high-quality and valuable products. This way the thesis challenges the conventional perception of material value and demonstrates a new design approach where imperfections and a green narrative are valued.

Further, the use of design thinking models such as the Double Diamond and reflective practice provides a methodological yet adaptive approach that can be used or reused by other designers. The research further identifies the need in B2B furniture environments to span emotional and functional value, a position increasingly important as firms aim to convey ethics and sustainability with their interior spaces.

Designers working in sustainable product design or office design can take the design process, user-centered design, and narrative-based strategy developed in this thesis forward. As sustainability increasingly plays a more important role in the future, this thesis can serve both as a point of inspiration and launching point for further research into the way waste materials can find new life through purposeful design intentions.

6.2 Conclusions and future scope

This thesis aimed to examine how a deserving lounge table should be designed regarding adopting the design values of desirability, viability, sustainability, and feasibility. The resulting design is not only a furniture piece and décor that will meet functional requirements, but also something that addresses an acute cause bark beetle infection causing wood wastage to Swedish forests. Taking infected woods to produce opulent, high-statement furniture thus determines how eco-design should remain sensible while being both strong and effective in its response to sustainability.

The design process was geared towards minimalism, local production, and storytelling, all of which contributed to making the table potentially marketable and desirable. Besides, feasibility was realized through strategic material selection, CNC machining, and joinery techniques optimized for available resources. Viability was also supported by linking the product to broader values of environmental consciousness and corporate ethics.

But while the product is well positioned at the idea level, the real test that remains is market validation. Until the product is launched and how people act is observed, its long-term sustainability and acceptance remain theoretical.

6.2.1 Future scope

Perhaps the most critical area for future development in this design is the concept of harmony, be it aesthetic or functional. While the current product successfully incorporates organic materials and an engaging storyline, there is room to investigate deeper the concept of harmony between design elements, materials, and intended setting. Understand what exactly makes a product so harmonious in its environment, by way of material contrasts, color schemes, or its positioning with other furniture pieces, can unlock new ways of innovation and fineness.

Additional research could further explore how the different aspects of design can be made to harmonize to make the product an even better fit in any setting. This would involve:

- Material Harmony: Investigating how different materials, such as wood, glass, and metal, work together in the context of this product could contribute to the overall visual and haptic experience. This research could also investigate how different surface treatments and textures could influence the sense of harmony in the product.
- Environmental and Cultural Congruity: There lies scope for study on how the lounge table relates to its settings, such as corporate spaces, hotels, or homes. How can the product blend with varied settings without diluting its own identity? Is it possible that some design shifts can be provided so they blend well with various interior decor schemes?
- Emotional Harmony: Beyond the physical setting, the emotional connection between product and user is also important. Investigating how the product's design can elicit positive emotional responses, such as calmness, inspiration, or a sense of oneness with nature, would result in a more balanced user experience.

Exploring these aspects of harmony can not only elevate the design stake but also expand the product's usefulness in a variety of situations, making it a more flexible and universally desirable thing. From the knowledge that increased comprehension of harmony in design makes possible, subsequent iterations of this product or of other such products could reach new levels of popularity, both in terms of consumer satisfaction and broad market appeal.

Subsequent research must entail the testing of the created lounge table in real usage environments. If time permitted, the writer could have performed the product testing by way of formal evaluations, such as moderator-conducted focus groups, and by direct contact with interior designers and office managers. These

groups were selected due to their demographic similarities to the most likely decision makers of office furniture acquisitions.

Conducting this kind of qualitative research would gain valuable perspective on the actual effect of the product, both in terms of functional application and imagined emotional and aesthetic worth. It would be a key step to ascertain whether the product was fulfilling its function and intended consumer base.

6.2.2 Validation considerations

Validation is a crucial stage in establishing the quality, authenticity, and dependability of the design. During this project, we managed to construct lounge tables. This allowed proportions, material texture and build quality to be tested early. However, as the deadline was restrictive and stakeholders weren't always available during the thesis period, more in-depth testing with real users wasn't feasible.

To absolutely validate the design, further stages are recommended. These include obtaining feedback from individuals who represent the target personas such as designers and office managers. Activities such as focus groups, contextual interviews and official usability testing would provide feedback related to the performance of the product in a real situation, the extent to which the narrative is being understood and the way form and function are perceived.

Validation also directly concerns the reliability of the outcome. By testing the generalizability and consistency of the product across numerous various stakeholders and varied settings, this validates the product to function as intended in a laboratory setup but also in everyday life.

If time allowed, it would have benefited to have conducted validation sessions where the users could have used the prototype and received detailed feedback. Having the ability to speak face-to-face with professional representatives of the target personas would have helped to ensure the table communicates its point as intended and is meeting both the intended functions and emotional expectations.

Though this stage affirms the concept is physically workable and tuned to the target design parameters, further user testing needs to be completed to evaluate its marketability and ultimate impact.

7 Reference list

[1] H. Petersson, "Skogsdata 2020," SLU institutionen för skoglig resurshållning, 2020. Accessed: Jan. 28, 2025. [Online]. Available: https://www.slu.se/globalassets/ew/org/centrb/rt/dokument/skogsdata/skogsdata_2020_webb.pdf

[2] F. Ekman, "Stormskadornas ekonomiska konsekvenser – hur ser försäkringsersättningsnivåerna ut inom familjeskogsbruket?," Pdf, 2009. Accessed: Jan. 28, 2025. [Online]. Available: https://stud.epsilon.slu.se/290/1/Exjobb_31.pdf

[3] "Vad gör ni av granbarkborretimret? - Vida AB," *Vida AB*, Nov. 27, 2020. https://www.vida.se/topical/vad-gor-ni-av-granbarkborretimret/ (accessed Jan. 28, 2025).

[4] "Lägre betalt för barkborreskadat virke," *Skogssallskapet.se*, Mar. 07, 2019. https://www.skogssallskapet.se/kunskapsbank/artiklar/2019-03-07-lagre-betalt-for-barkborreskadat-virke.html (accessed Jan. 29, 2025)

[5] D. A. Schön, *The reflective practitioner: how professionals think in action*. New York: Basic Books, 1983.

[6] S. Sinek, *Start with why : how great leaders inspire everyone to take action*. New York: Portfolio/Penguin, 2009.

[7] S. Frey, J. B. Am, V. Doshi, A. Malik, and S. Noble, "Consumers Care about Sustainability—and Back It up with Their Wallets," *McKinsey & Company*, Feb. 06, 2023. https://www.mckinsey.com/industries/consumer-packaged-goods/our-insights/consumers-care-about-sustainability-and-back-it-up-with-their-wallets

[8] "Om oss - Sense by US AB," *Sense by US AB*, 2025. https://www.sensebyus.se/17/2/om-oss/ (accessed Jan. 29, 2025).

[9] K. T. Ulrich, S. D. Eppinger, and M. C. Yang, Product design and development, Seventh edition (international student edition). New York, NY: McGraw-Hill Education, 2020.

[10] S. Satir and D. G. Ertaş, *Furniture as a Design Product*, Jan. 2021. doi: https://www.researchgate.net/publication/359843744_Furniture_as_a_Design_Product

[11] T. Yu Siang, "What is Design Thinking?," *The Interaction Design Foundation*, 2019. https://www.interaction-design.org/literature/topics/design-thinking

[12] British Design Council, "The Double Diamond," *www.designcouncil.org.uk*, 2015. https://www.designcouncil.org.uk/our-resources/the-double-diamond/

[13] M. Majka, "Mastering Product Development with the Double Diamond Framework,", Oct. 2024. https://www.researchgate.net/publication/384691492_Mastering_Product_Development_with_the_Double_Diamond_Framework (accessed Feb. 02, 2025).

[14] B. Lawson, *How designers think : the design process demystified*, 4. ed. Oxford: Architectural Press, 2006.

[15] S. Nordgren och J. Sällberg, *Personakt Sven Lundh : en bok om Källemos skapare*. Malmö: Arena, 2007.

[16] S. Sinek, "The golden circle," *Simon Sinek*, 2009. https://simonsinek.com/golden-circle/

[17] A. Warell, *Modelling perceptual product experience – Towards a cohesive framework of presentation and representation in design*, 2008.

[18] D. A. Norman, *Emotional design: why we love (or hate) everyday things*. New York: Basic Books, 2004.

[19] S. Gibbons, "Empathy mapping: the first step in design thinking," *Nielsen Norman Group*, Jan. 14, 2018. https://www.nngroup.com/articles/empathy-mapping/

[20] R. Elmansy, "The Double Diamond Design Thinking Process and How to Use it," *Designorate*, Feb. 09, 2021. https://www.designorate.com/the-double-diamond-design-thinking-process-and-how-to-use-it/

[21] A. Bhat, "Market Research: What it Is, Methods, Types & Examples," *QuestionPro*, Apr. 27, 2018. https://www.questionpro.com/blog/what-is-market- research/

[22] R. F. Dam and Teo Yu Siang, "Personas – A Simple Introduction," *The Interaction Design Foundation*, Nov. 17, 2016. https://www.interaction-design.org/literature/article/personas-why-and-how-you-should-use-them?srsltid=AfmBOoqJWw83jyFQcE8B-Y7yjvQ7jk5Ygg9Vtrqi52wDb_JNyRZffLP5 (accessed feb. 12, 2025).

[23] K. D. on, "What is a Mood Board — Definition, Examples & How They Work," *StudioBinder*, Jan. 17, 2020. https://www.studiobinder.com/blog/what-is-a-mood-board-definition/

[24] "Requirements vs. Specifications," *ArgonDigital | Making Technology a Strategic Advantage*, Oct. 25, 2021. https://argondigital.com/blog/product-management/requirements-vs-specifications-create-a-shared-vocabulary/

[25] S. Doorley, S. Holcomb, P. Klebahn, J. Utley, and K. Segovia, "Design thinking bootleg," Hasso Plattner Institute of Design at Stanford, 2018. Accessed: Feb. 12, 2025. [Online]. Available: https://dschool.sfo3.digitaloceanspaces.com/documents/dschool_bootleg_deck_2018_final_sm2-6.pdf

[26] A. Jangra, "What is a Mockup? Definition, Examples & FAQ - Supahub," *supahub.com*, Mar. 02, 2024. https://supahub.com/glossary/what-is-a-mockup (accessed Feb. 13, 2025).

[27] M. Segasby, "What is Product Rendering?," *www.lightmap.co.uk*, Sep. 20, 2022. https://www.lightmap.co.uk/blog/product-rendering/ (accessed Feb. 13, 2025).